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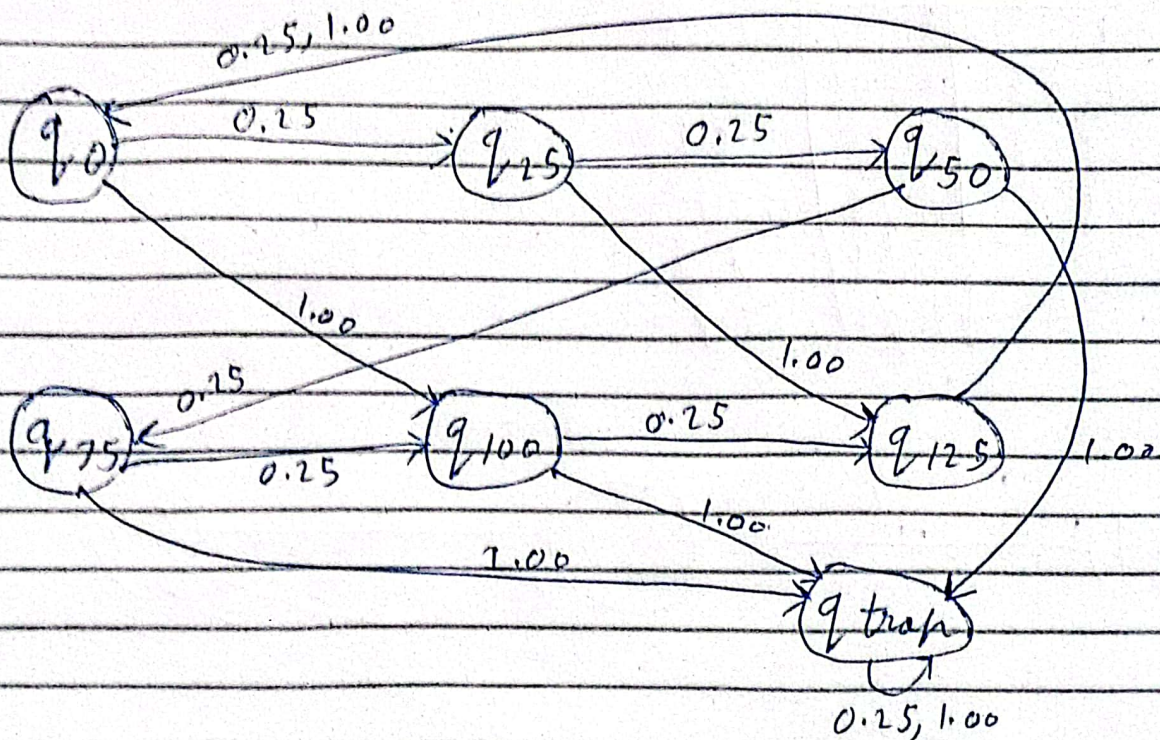
Subject: Theory of Automata (B)

Sap ID# 70150006

Assignment #2

Q₄ Construct a DFA for the vending machine that accept coin of of \$0.25 and \$1.00. You get coke drink from this machine with \$1.25.

DFA



Q #24

Conversion of NFA into DFA

Soln

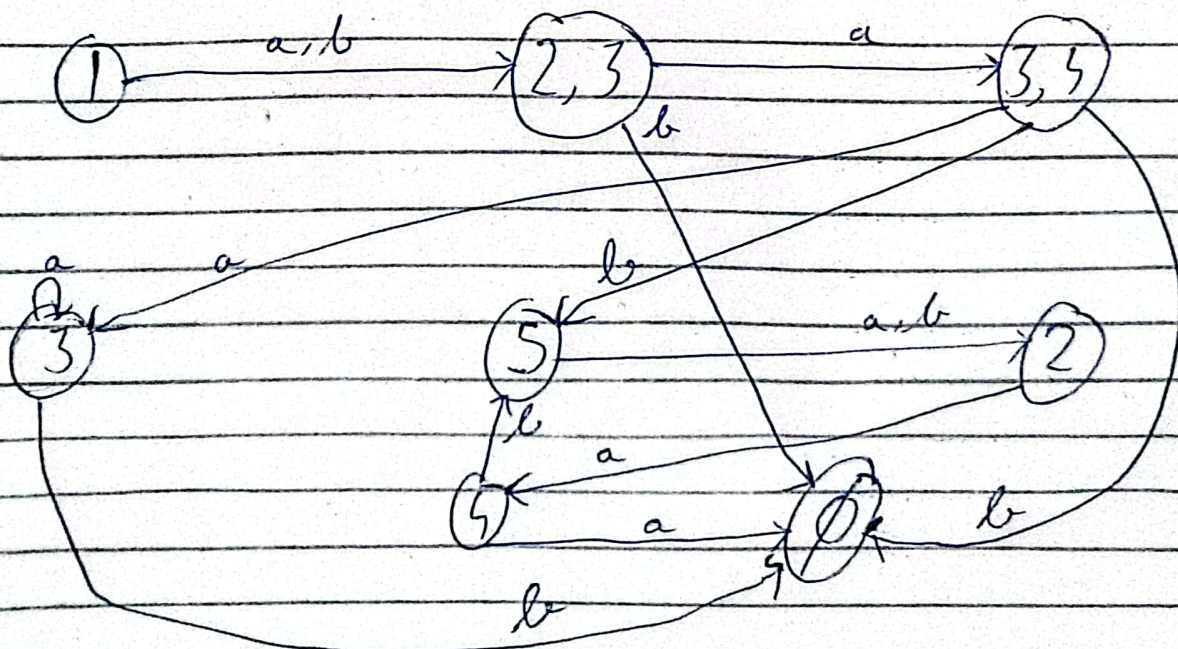
Transition table

	a	b
1	{2,3}	{2,3}
2	{4}	$\emptyset$
3	{3}	$\emptyset$
4	$\emptyset$	{5}
5	{2}	{2}

table # 2

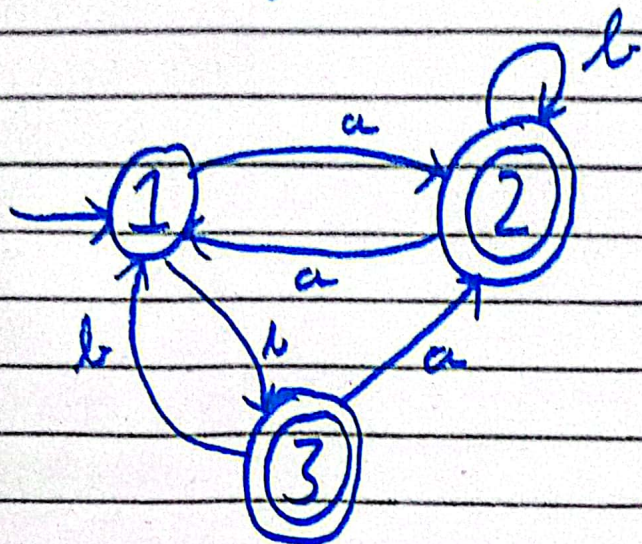
States	a	b
1	{2,3}	{2,3}
{2,3}	{3,4}	$\emptyset$
{3,4}	{3}	{5}
{5}	{2}	{2}
2	{4}	{ $\emptyset$ }
4	{ $\emptyset$ }	{5}

DFA

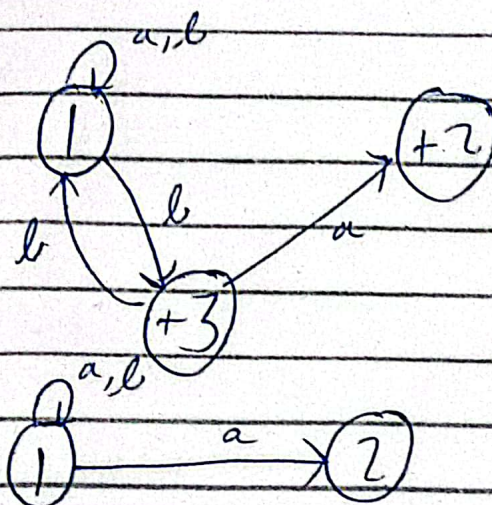


Q#3

Convert the following language into regular expression



Sol_n



R.E: $(a+b)^*a$